



Fundamentals of Optics

Lecturer

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Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

CHAPTER 5

MORE ON GEOMETRICAL OPTICS

□ The preceding chapter, there are two predominant approximations:

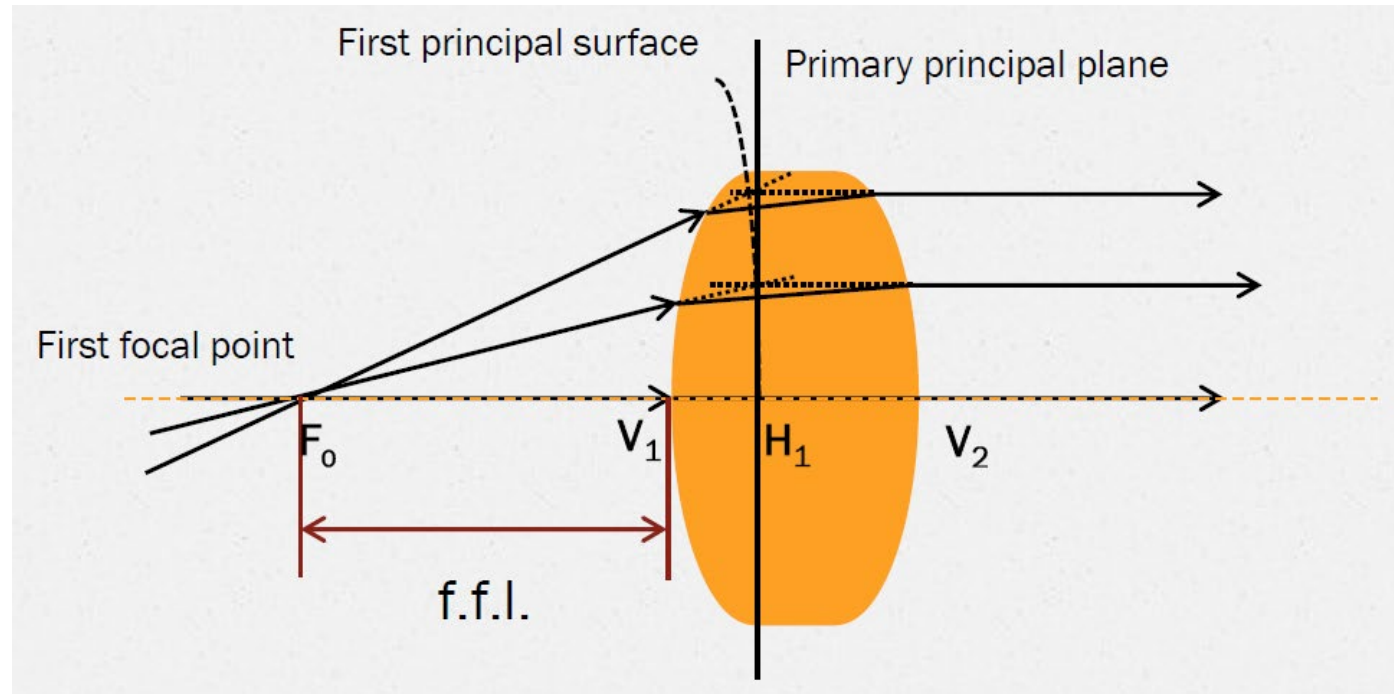
- ✓ Thin lenses
- ✓ First-order theory

□ For the design of a precision optical system:

- ✓ Thick lenses
- ✓ Aberrations

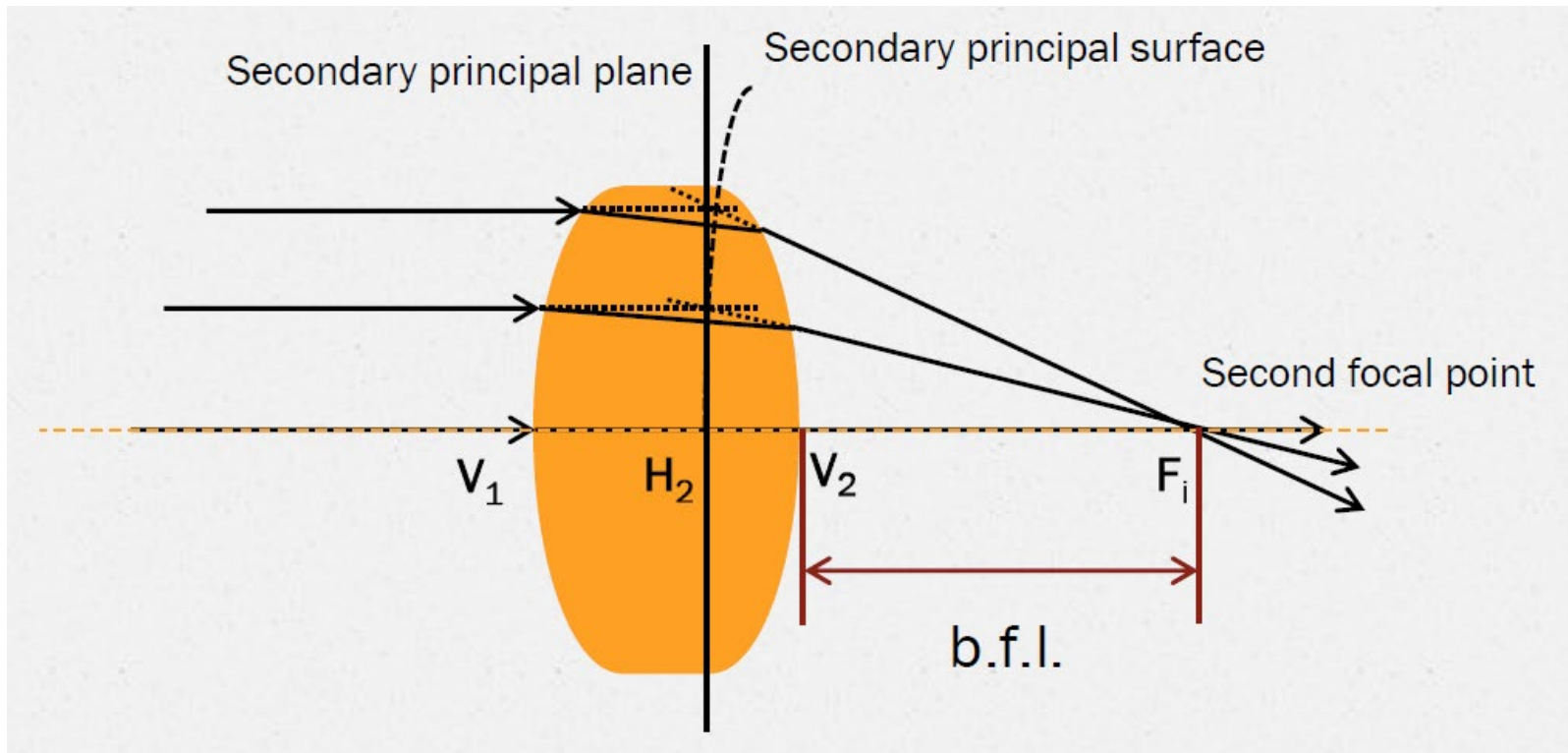
5.1 Thick Lenses and Lens Systems

Object-side (or first) principal surface: a single light ray, passing through the object-side focal point F_o , is refracted by both surfaces of the lens and exits parallel to the optical axis. Extensions of the refracted and incident ray intersect at a surface.



Primary principal planes: near the vicinity of the lens axis, the surfaces approximate plane surfaces.

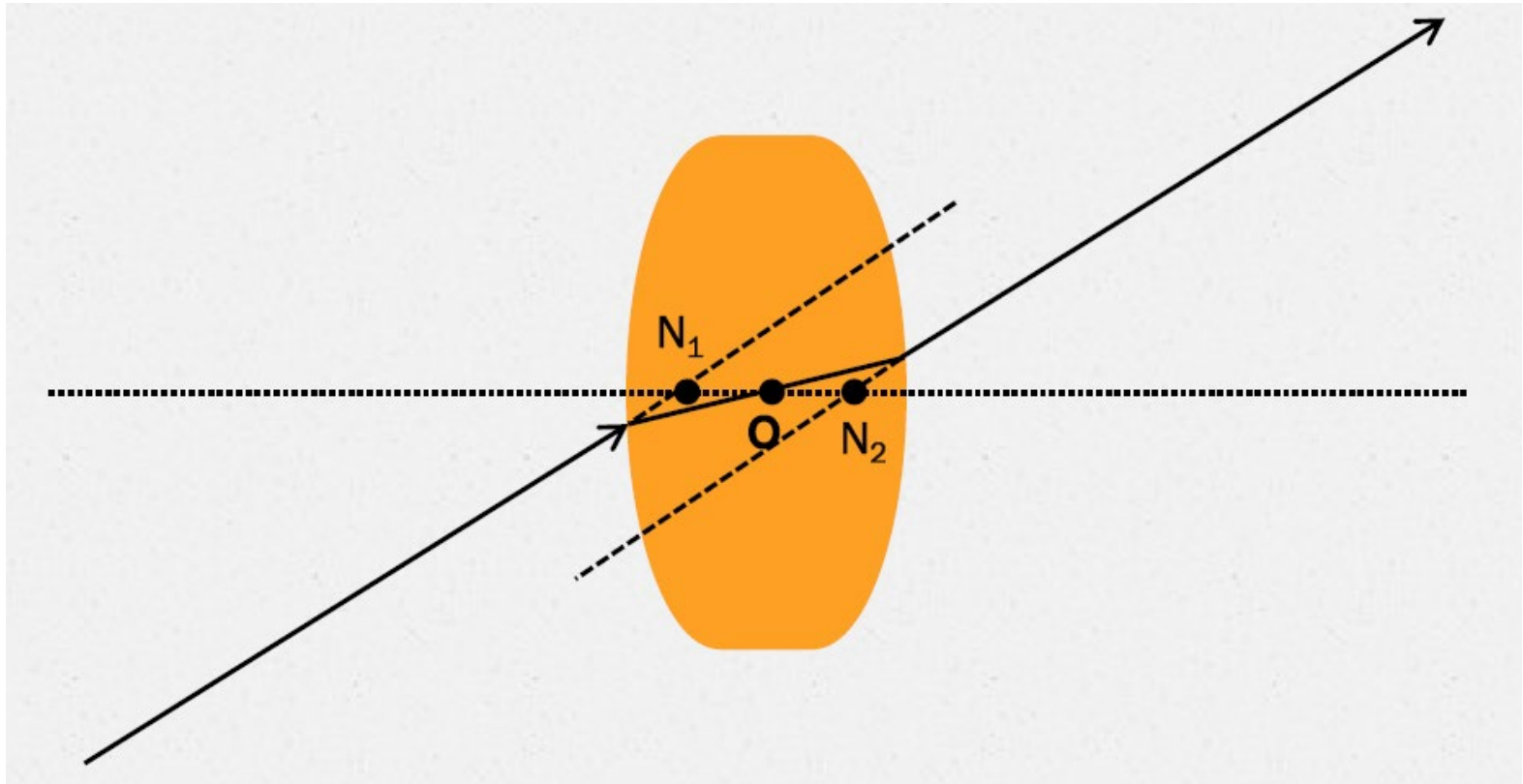
First principal points [H_1]: the primary principal planes intersect the optical axis.



Secondary principal planes: near the vicinity of the lens axis, the surfaces approximate plane surfaces.

Second principal points [H_2]: secondary principal planes intersect the optical axis.

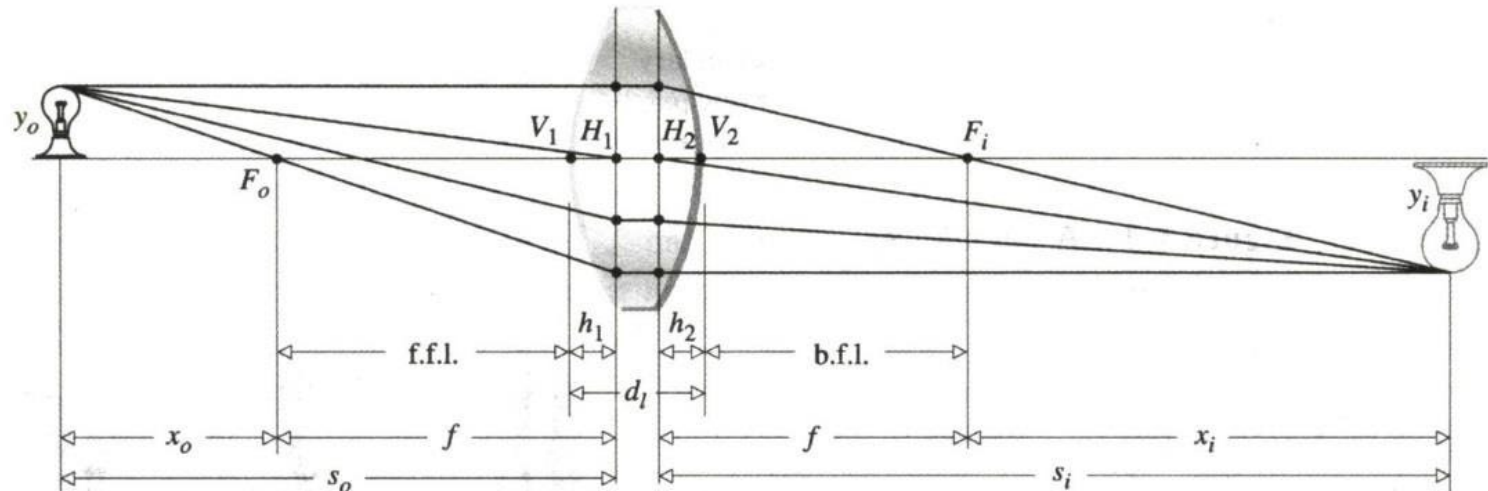
Earlier, we know that a ray traversing the lens through its optical center emerges parallel to the incident direction. Extending both the incoming and outgoing rays until they cross the optical axis located what are called the **nodal points**.



When the lens is surrounded on both sides by the same medium, generally air, the nodal and principal points will be coincident.

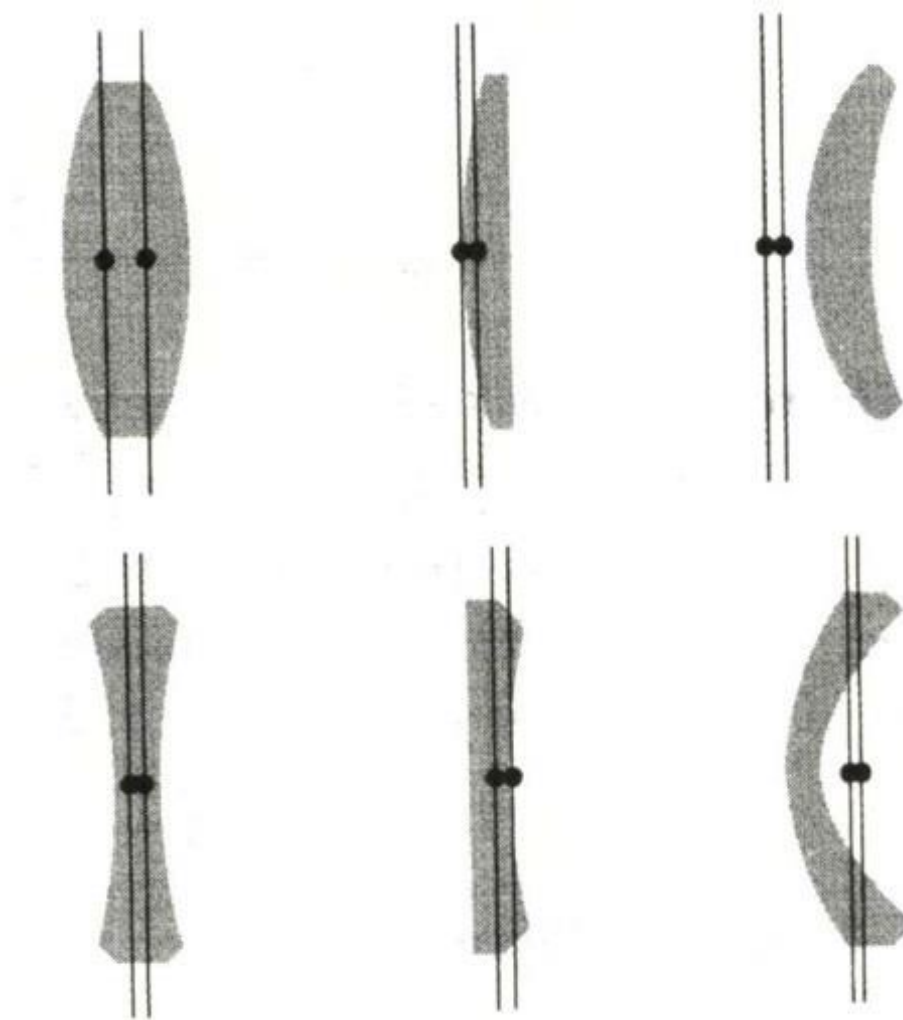
The six points, two focal, two principal, and two nodal, constitute the cardinal points of the system.

Function of principal planes: the refraction happened at the principal planes.

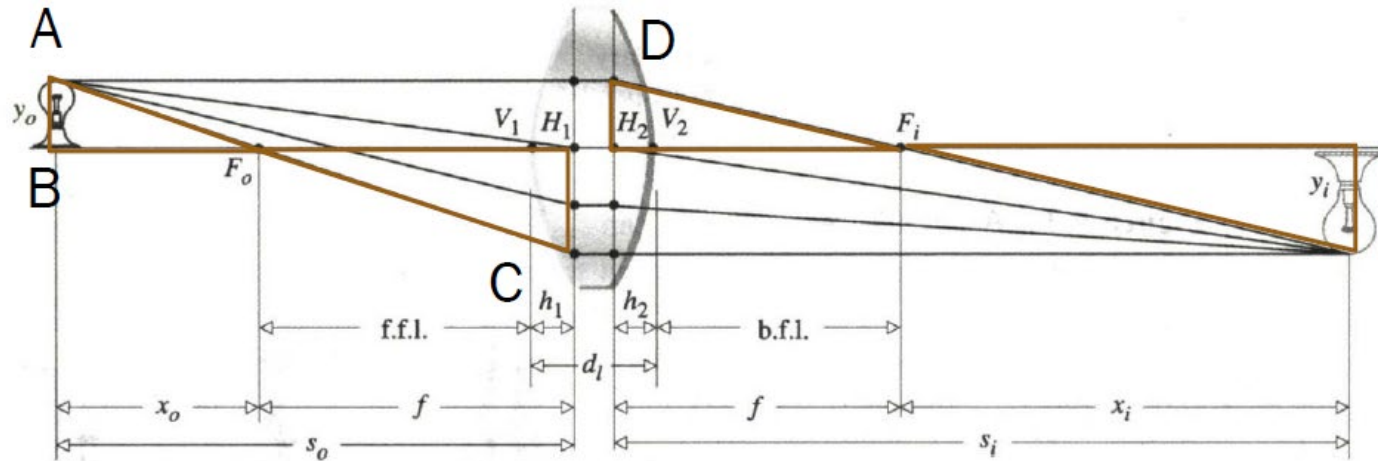


- ☐ An incident ray parallel to the main axis bends its direction at the second principal plane and passes through F_i .
- ☐ An incident ray passing through F_o bends its direction at the first principal plane and makes parallel to the main axis.
- ☐ An incident ray reaching point H_1 goes to H_2 and then propagates parallel to the original direction.

Principal plane for different lens bending



The thick lens is treated as consisting of two spherical refracting surfaces separated by a distance d_l between their vertices.



$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

$$M_T = \frac{y_i}{y_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$

s_o and s_i : the object and image distances measured from the first and second principal planes, respectively.

The effective focal length $\frac{1}{f} = (n_l - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n_l - 1)d_l}{n_l R_1 R_2} \right]$

$$h_1 = -\frac{f(n_l - 1)d_l}{n_l R_2}$$

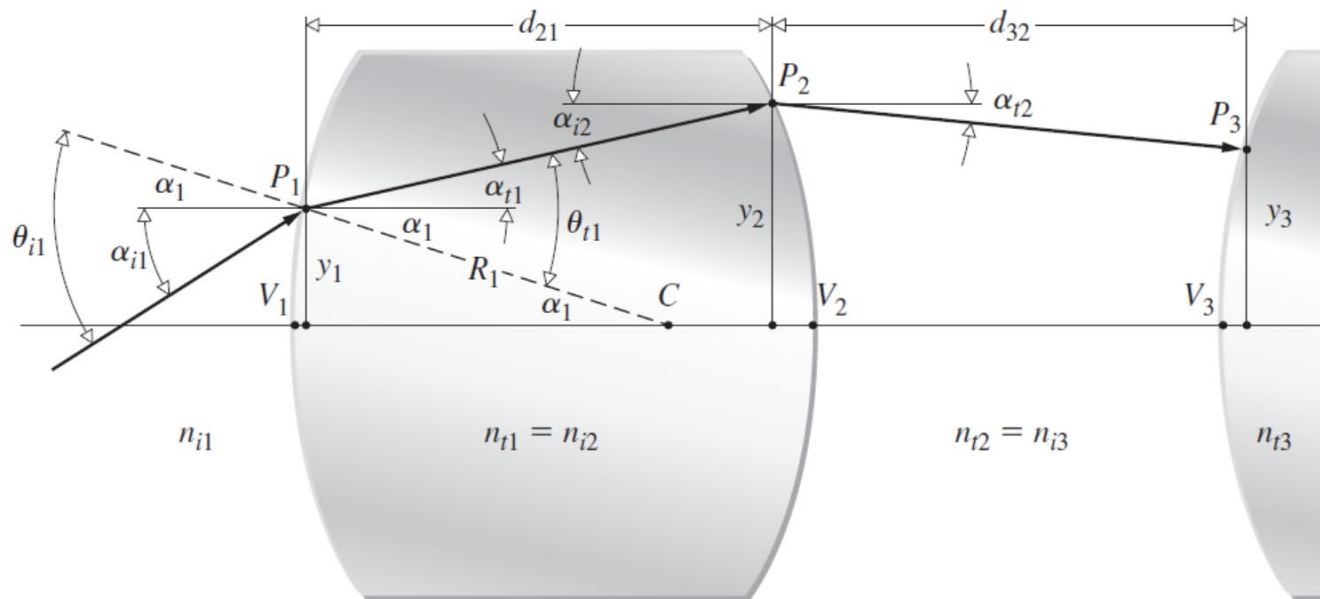
$$h_2 = -\frac{f(n_l - 1)d_l}{n_l R_1}$$

h_1 and h_2 are positive when the planes lie to the right of their respective vertices.

Analytical Ray Tracing

Method: Snell's law on all the surface

The simple case: a paraxial, meridional ray traveling a thick spherical lens.



Refraction equation: $n_{t1}\alpha_{t1} = n_{i1}\alpha_{i1} - \mathcal{D}_1 y_1$ $\mathcal{D}_1 = \frac{n_{t1} - n_{i1}}{R_1}$

Transfer equation: $y_2 = y_1 + d_{21}\alpha_{t1}$ $d_{21} \approx \overline{V_2 V_1}$

Matrix Analysis of Lenses

On the first surface

$$n_{t1}\alpha_{t1} = n_{i1}\alpha_{i1} - \mathcal{D}_1 y_{i1}$$

$$y_{t1} = 0 + y_{i1}$$

Matrix form

$$\begin{bmatrix} n_{t1}\alpha_{t1} \\ y_{t1} \end{bmatrix} = \begin{bmatrix} 1 & -\mathcal{D}_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} n_{i1}\alpha_{i1} \\ y_{i1} \end{bmatrix}$$

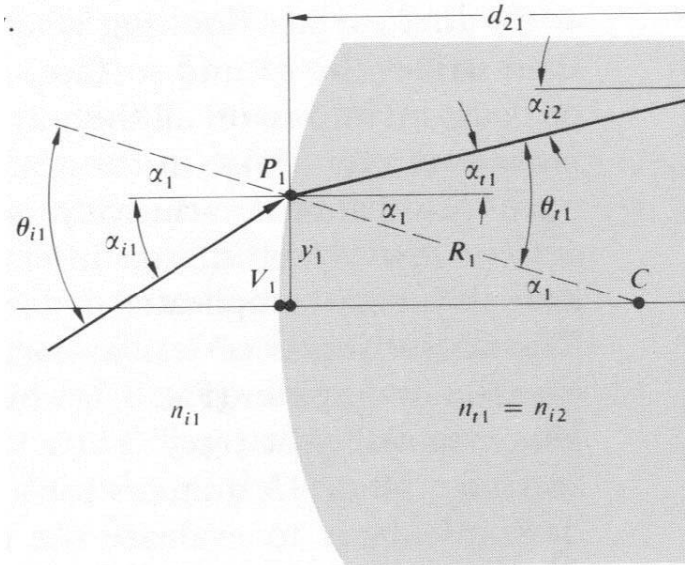
In any case, there can be envisioned as rays on either side of P1, one before and the other after refraction.

$$\mathcal{r}_{t1} \equiv \begin{bmatrix} n_{t1}\alpha_{t1} \\ y_{t1} \end{bmatrix} \quad \mathcal{r}_{i1} \equiv \begin{bmatrix} n_{i1}\alpha_{i1} \\ y_{i1} \end{bmatrix}$$

Refraction matrix

$$\mathcal{R}_1 = \begin{bmatrix} 1 & -\mathcal{D}_1 \\ 0 & 1 \end{bmatrix}$$

$$\mathcal{r}_{t1} = \mathcal{R}_1 \mathcal{r}_{i1}$$



From P_1 to P_2 :

$$n_{i2}\alpha_{i2} = n_{t1}\alpha_{t1} + 0$$

$$y_{i2} = d_{21}\alpha_{t1} + y_{t1}$$

Matrix form

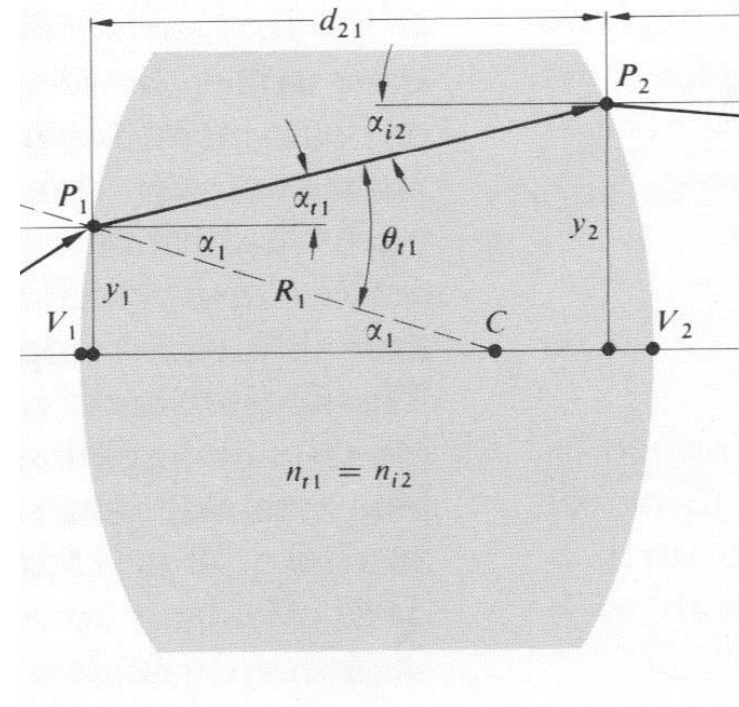
$$\begin{bmatrix} n_{i2}\alpha_{i2} \\ y_{i2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ d_{21}/n_{t1} & 1 \end{bmatrix} \begin{bmatrix} n_{t1}\alpha_{t1} \\ y_{t1} \end{bmatrix}$$

Transfer matrix

$$\mathcal{T}_{21} = \begin{bmatrix} 1 & 0 \\ d_{21}/n_{t1} & 1 \end{bmatrix}$$

$$\mathcal{r}_{i2} \equiv \begin{bmatrix} n_{i2}\alpha_{i2} \\ y_{i2} \end{bmatrix}$$

$$\mathcal{r}_{i2} = \mathcal{T}_{21}\mathcal{r}_{t1} = \mathcal{T}_{21}\mathcal{R}_1 \mathcal{r}_{i1}$$



On the second surface

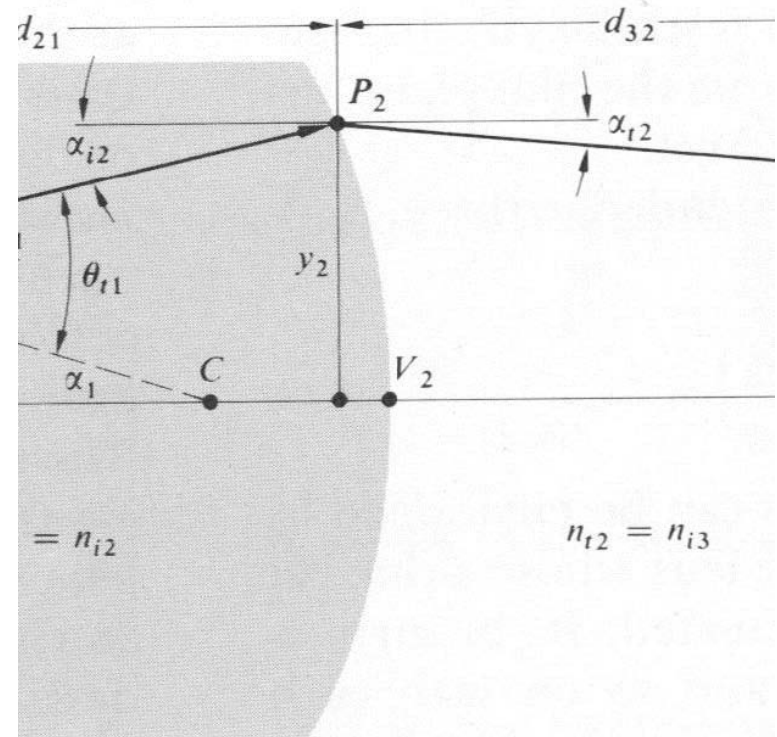
$$\mathcal{R}_2 = \begin{bmatrix} 1 & -\mathcal{D}_2 \\ 0 & 1 \end{bmatrix}$$

where $\mathcal{D}_2 = \frac{n_{t2} - n_{i2}}{R_2}$

$$\mathbf{r}_{t2} = \mathcal{R}_2 \mathbf{r}_{i2}$$

where $\mathbf{r}_{t2} = \begin{bmatrix} n_{t2} \alpha_{t2} \\ y_{t2} \end{bmatrix}$

$$\mathbf{r}_{t2} = \mathcal{R}_2 \mathcal{T}_{21} \mathcal{R}_1 \mathbf{r}_{i1}$$



$$r_{t2} = \mathcal{R}_2 \mathcal{T}_{21} \mathcal{R}_1 r_{i1}$$

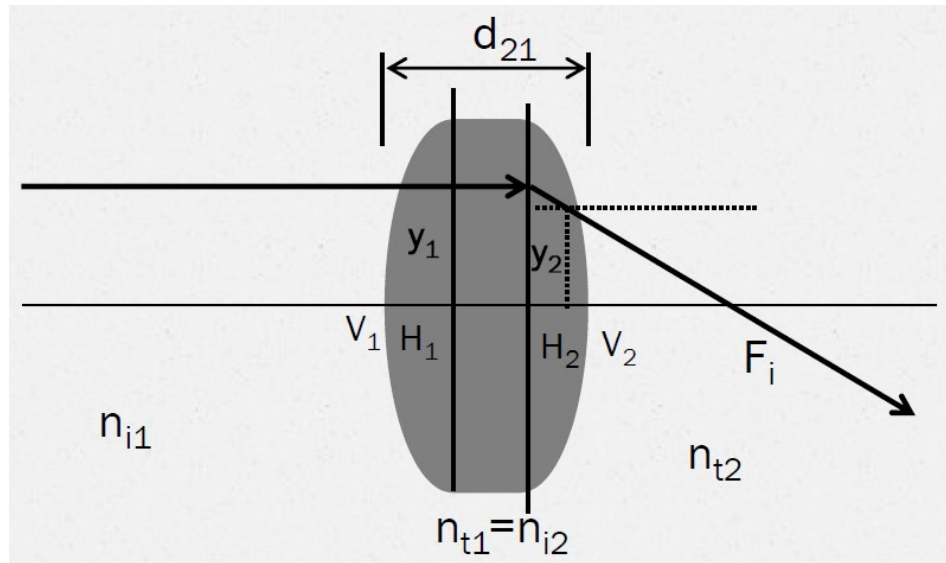
System matrix $\mathcal{A}$: $\mathcal{A} \equiv \mathcal{R}_2 \mathcal{T}_{21} \mathcal{R}_1$

$$\begin{aligned} \mathcal{A} &\equiv \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & -\mathcal{D}_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ d_{21}/n_{t1} & 1 \end{bmatrix} \begin{bmatrix} 1 & -\mathcal{D}_1 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 - \frac{\mathcal{D}_2 d_{21}}{n_{t1}} & -\mathcal{D}_1 - \mathcal{D}_2 + \frac{\mathcal{D}_1 \mathcal{D}_2 d_{21}}{n_{t1}} \\ \frac{d_{21}}{n_{t1}} & 1 - \frac{\mathcal{D}_1 d_{21}}{n_{t1}} \end{bmatrix} = \begin{bmatrix} 1 - \frac{\mathcal{D}_2 d_l}{n_l} & -\mathcal{D}_1 - \mathcal{D}_2 + \frac{\mathcal{D}_1 \mathcal{D}_2 d_l}{n_l} \\ \frac{d_{21}}{n_l} & 1 - \frac{\mathcal{D}_1 d_{21}}{n_l} \end{bmatrix} \end{aligned}$$

The system matrix $\mathcal{A}$ in this case transforms an incident ray at the first surface to an emerging ray at the second surface.

$$\boxed{\begin{bmatrix} n_{t2} \alpha_{t2} \\ y_{t2} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} n_{i1} \alpha_{i1} \\ y_{i1} \end{bmatrix}}$$

Effective focal length and principal point location



Consider the upper part of ray first, where $\alpha_{i1}=0$

$$\begin{bmatrix} n_{t2}\alpha_{t2} \\ y_{t2} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} n_{i1}\alpha_{i1} \\ y_{i1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 0 \\ y_{i1} \end{bmatrix} = \begin{bmatrix} a_{12}y_{i1} \\ a_{22}y_{i1} \end{bmatrix}$$

The second focal length f_2 ($\alpha_{t2} < 0$):

$$f_2 = \overline{H_2 F_i} = \frac{y_{i1}}{-\alpha_{t2}} = \frac{y_{i1}}{-a_{12}y_{i1}/n_{t2}} = -\frac{n_{t2}}{a_{12}}$$

$$\overline{H_2 V_2} = \frac{y_{i1} - y_{t2}}{-\alpha_{t2}} = \frac{y_{i1} - a_{22}y_{i1}}{-a_{12}y_{i1}/n_{t2}} = -\frac{(a_{22} - 1)n_{t2}}{a_{12}} = -\frac{f_2(n_{t2} - 1)}{n_{t2}}$$